



University of Colombo, Sri Lanka

University of Colombo School of Computing

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

Third Year Examination - Semester I -UCSC AY20 [held in October 2023]

SCS3204 - Management

(Two (2) Hours)

Answer ALL questions

Number of Pages = 16

Number of Questions = 4



To be completed by the candidate

Index Number:

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Important Instructions to candidates:

- I. Students should answer in the medium of English language only using the space provided in this question paper.
- II. Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
- III. Write your index number CLEARLY on each and every page of this Question paper.
- IV. This paper consists of 4 questions in 16 pages (including the Cover Page).
- V. Answer ALL questions.
- VI. Programmable Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are not allowed.
- VII. Non-programmable Calculators are allowed.
- VIII. Do not tear off any part of this answer book. Under no circumstances may this book, used or unused, be removed from the Examination Hall by a candidate.

To be completed by the examiners

1	
2	
3	
4	
Total	

Index Number:

Question 01

Read the following case study and answer the questions given below.

“GRANTHA” is a bookshop that has books of all kinds. It also has Note Pads, Paper products, Coloring products, school stationaries, office stationaries, novels (English/Sinhala/Tamil), etc.

GRANTHA opened six years ago in Nugegoda which is closer to a government university, private university and two schools. There are four (4) other leading bookshops in and around that area. GRANTHA is owned by a family of scholars who owns a publishing company which has been producing books for Sri Lanka for the past 15 years. GRANTHA is the first shop they opened and it is doing very well in the area where books and accessories are being used extensively.

GRANTHA also has partnerships and is the sole distributor for a world-renowned brand of accessories as well. Their factories produce all the paper-based products sold at GRANTHA and they have the latest technology guarantying the quality of the brand being sold. They follow the 5S standards and has been awarded the ISO 9000 certification as well.

GRANTHA brand has won many awards for its superior quality products, and currently, it's the 3rd leading book manufacturer in Sri Lanka. Book industry is very competitive in Sri Lanka. In every corner of any street, there is at least a small book shop. There are importers of books at very competitive prices as well as small scale manufacturers and printers as well.

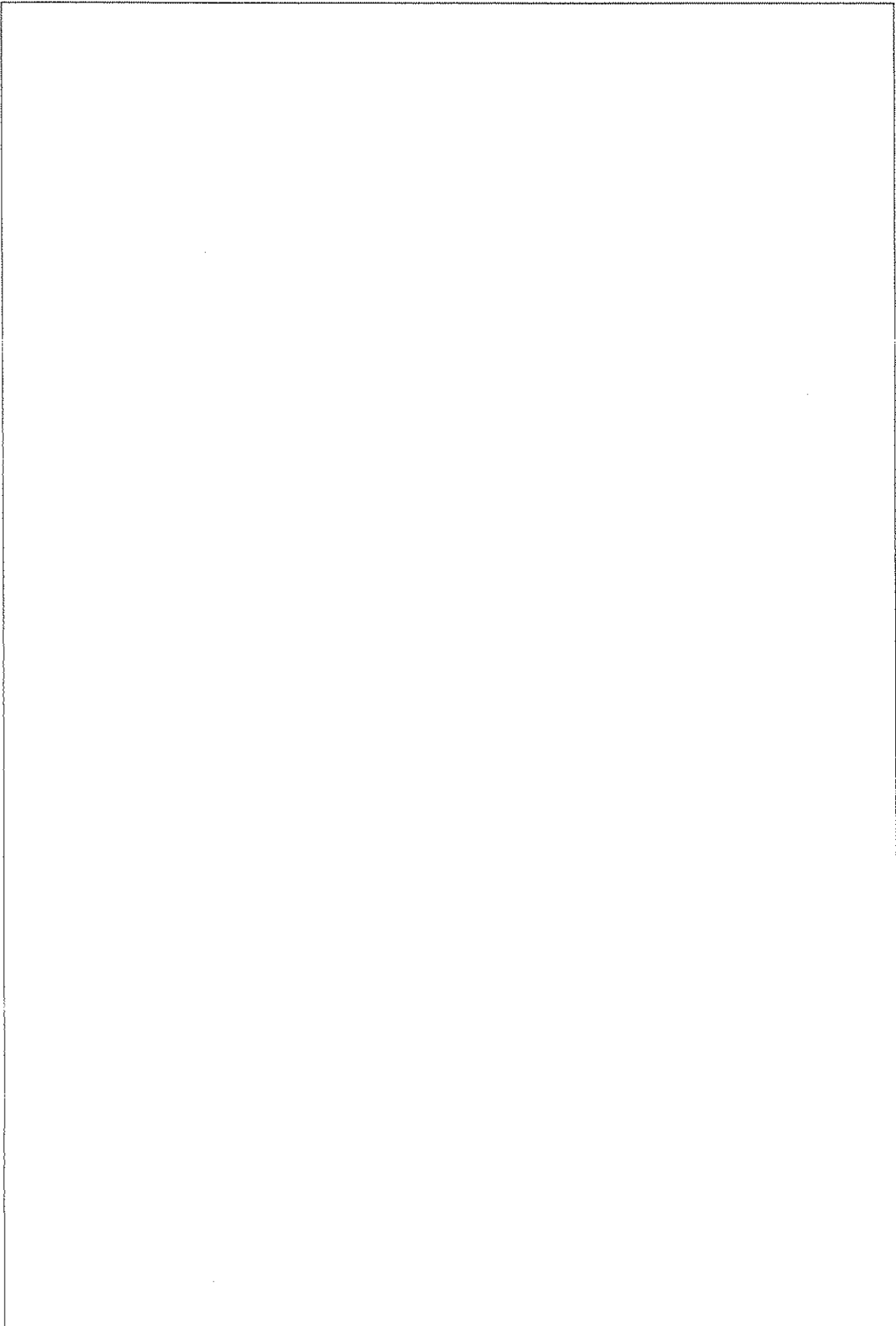
Paper and other material needed such as ink and plastics are imported and other raw materials are bought from few local buyers. The price fluctuations of these very frequently affect the pricing models of GRANTHA. Shortages of paper and price increases in raw materials result from the economic crisis in Sri Lanka. The company markets the shop and products using leaflets and posters in the area it is situated in.

GRANTHA bookshop wants to expand their business into online selling of books since a need was created for online transactions after the Covid-19 pandemic. They want to expand in to selling educational software and CDs as well. GRANTHA is also looking at partnerships and collaborations with related service providers to expand their business island wide.

- (a) GRANTHA wants to introduce new services to the market to expand their business. Analyze the existing company situation and the potential of the company by carrying out a SWOT Analysis.

(10 Marks)

Index Number:



Index Number:

- (b) Use Michael Porter's Five (5) Forces Analysis to identify and understand the competitive forces working on GRANTHA. Justify your answer by explaining why you think certain forces are high and why certain forces are medium or low.

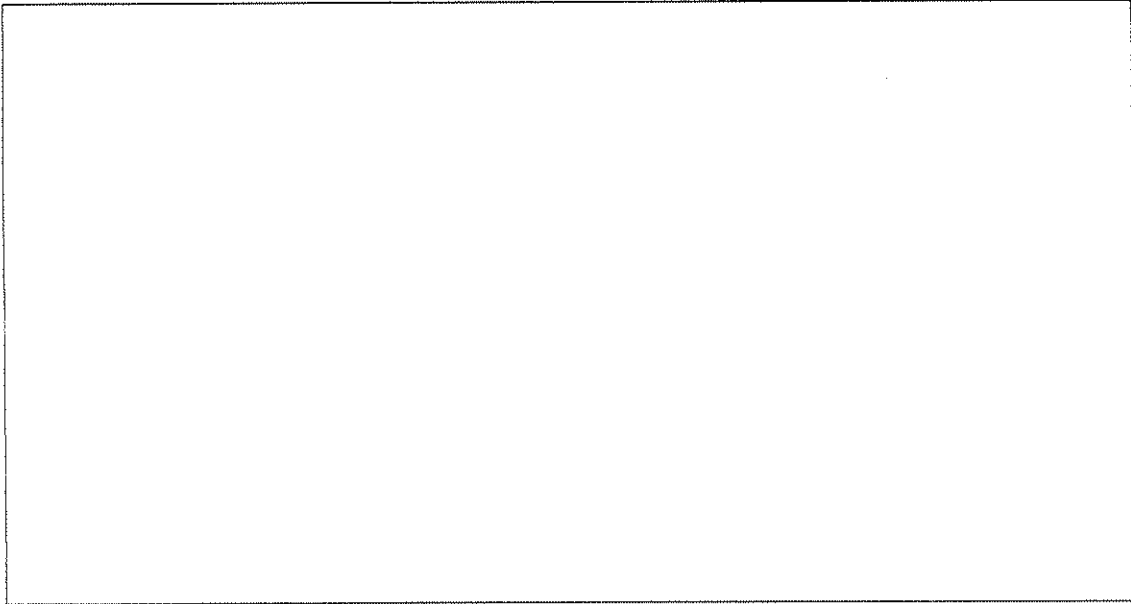
(9 Marks)

- (c) Propose 2 strategies for the company GRANTHA to overcome the forces discussed under question 1 (b). Justify your answer and explain how these strategies would help the company. (6 Marks)

Question 02

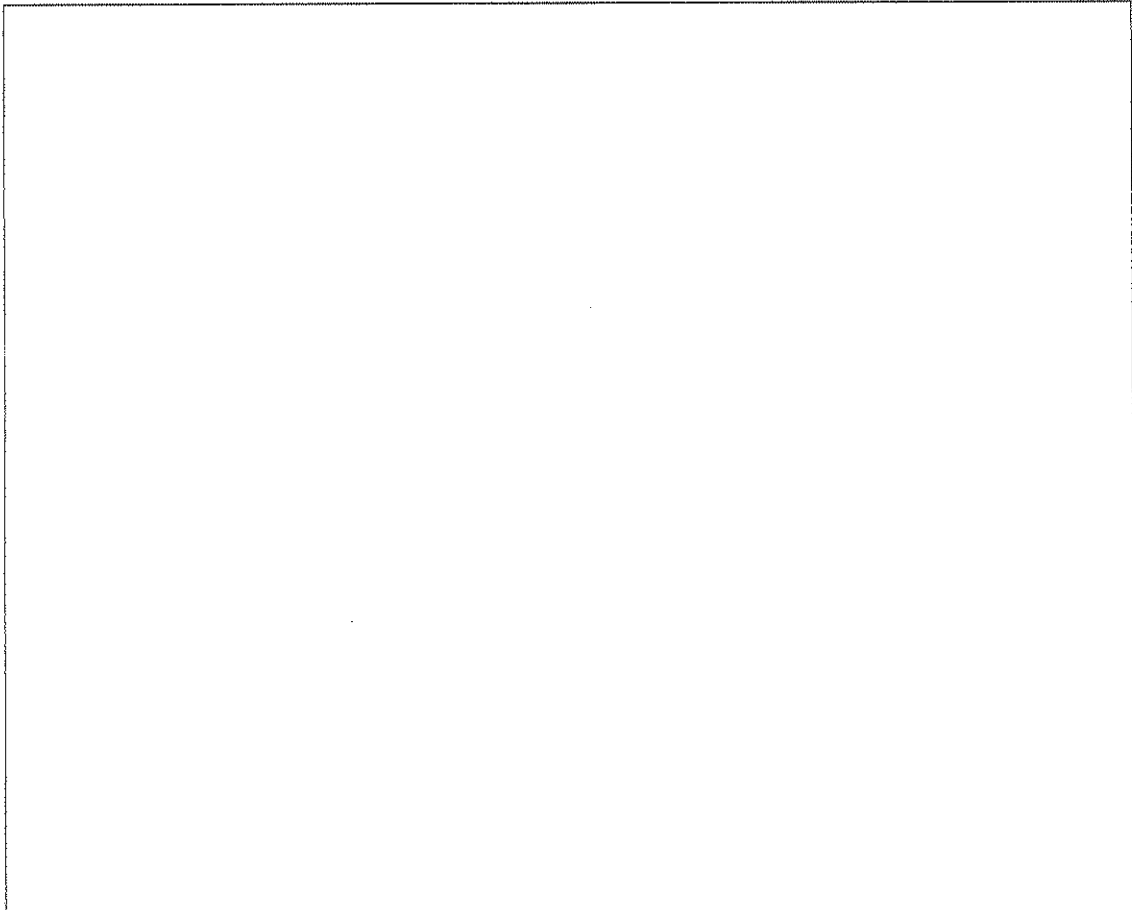
- (a) Write down a brief description for the following with an example each.
- i. The Mom Test in Ethical Decision-making.
 - ii. The TV Test in Ethical Decision-making.
 - iii. Actions that are Legal and Not Ethical.
 - iv. Actions that are Ethical and Not Legal.
- (8 Marks)

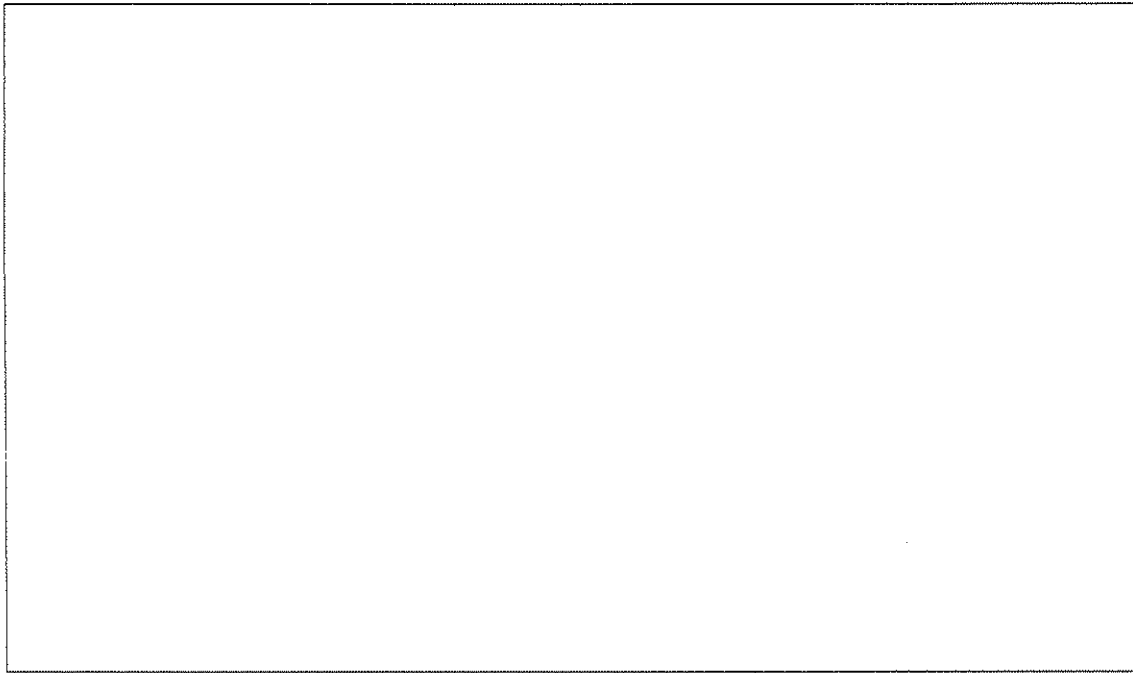
Index Number:



- (b) A well known Computer Hardware seller is hoping to introduce and sell Tabs to school children of O/L and A/L classes, for educational purposes. Develop a Marketing Strategy to sell this new product to the target market. Justify your answer and explain why these particular strategies are proposed.

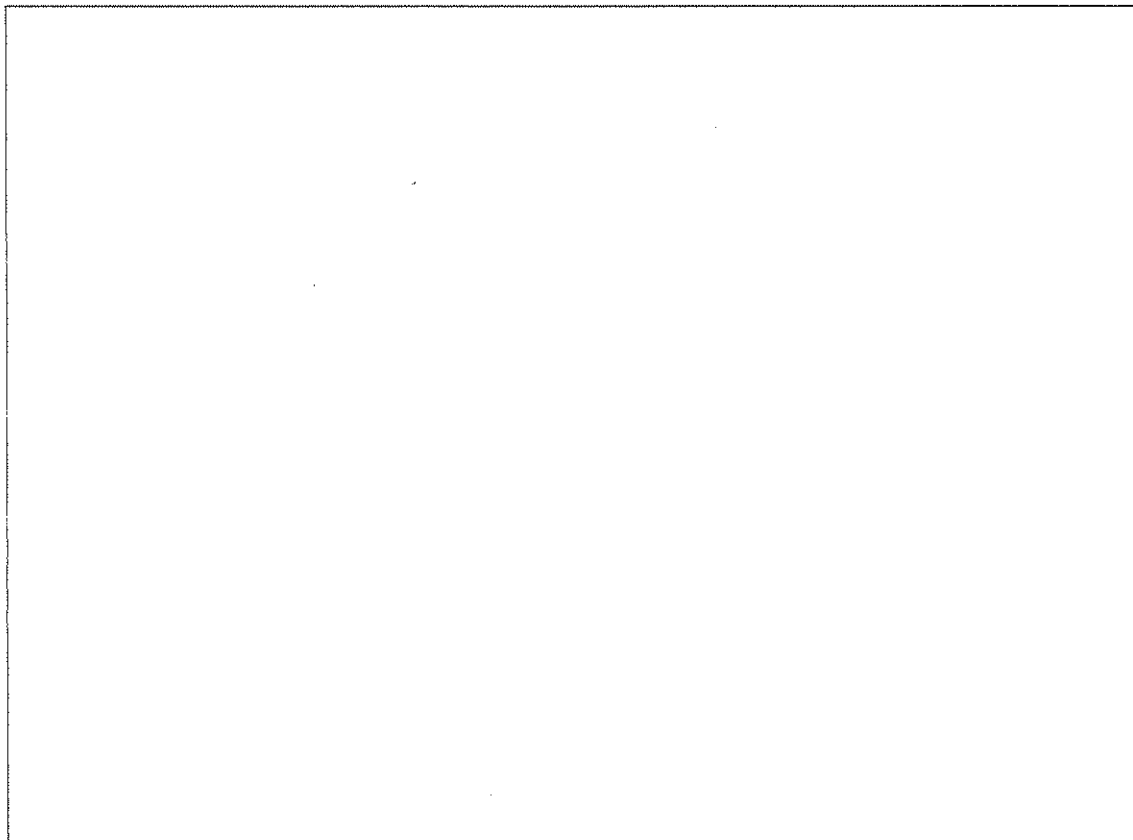
(12 Marks)





- (c) Product digitization has made a revolutionary change in Business. Briefly explain five (5) products that can be digitized to bring revenue to the business and five (5) products or services that cannot be so digitized.

(05 Marks)



Index Number:

Question 03

- a) The following Statement of Financial Position has been prepared for Ruhunu Enterprises as at 31/12/2021.

Ruhunu Enterprises
Statement of Financial Position
As at 31/12/2021

	Rs.	Rs.
<u>Non-Current Assets</u>		
Land & Buildings	150,000.00	
Machinery	100,000.00	250,000.00
<u>Current Assets</u>		
Cash	65,000.00	
Bank	40,000.00	
Stocks	12,000.00	
Debtors	13,000.00	130,000.00
		380,000.00
<u>Equity & Liabilities</u>		
Capital		250,000.00
<u>Non-Current Liabilities</u>		
Bank Loan		100,000.00
<u>Current Liabilities</u>		
Creditors		30,000.00
		380,000.00

The following transactions are taken place during the year 2022.

- Total purchases during the period
 - On credit: Rs. 60,000.00
 - On cash: Rs. 40, 000.00
- Total Sales during the period
 - On credit: Rs. 80,000.00
 - On cash: Rs. 60, 000.00
- Paid electricity bill of Rs. 3,000.00
- Rs. 20,000.00 worth of a cheque received from debtors.
- Invested extra capital with of Rs. 20,000.00

Additional Information,

Machinery Depreciated at 10% annually.

Closing stock as 31/12/2022 was Rs. 8,000.00

- i. Prepare the Statement of Profit & Loss (Income Statement) for the year ended 31/12/2022.

[08 Marks]

Index Number:

- ii. Prepare the Statement of Financial Position (Balance sheet) as at 31/12/2022.

[11 Marks]

b) List down 02 primary functions of the accounting process.

[02 Marks]

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c) Write down 01 essential characteristic of Assets?

[02 Marks]

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d) Make the journal entry for the following transaction.

- Credit sales Rs. 32,000 on 15th of September, 2023.

[02 Marks]

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Index Number:

Question 04

- a) The following Information has been extracted from the financial statements of Amalka Ltd for the years ending 2021 and 2022.

	2021	2022
Sales	100,000	125,000
Gross Profit	40,000	60,000
Net Profit	21,000	30,000
Income Tax Expenses	1,500	2,000
Current Assets	50,000	80,000
Non-Current Assets	100,000	120,000
Current Liabilities	50,000	40,000
Non- current Liabilities	100,000	150,000
Equity	225,000	300000

You are requested to calculate the following financial ratios using the provided data for the years 2021 and 2022. And provide a brief analysis of the organization's performance in 2022 in comparison to 2021 using the calculated ratios.

[09 Marks]

- Return on Assets

- Current Ratio

- Equity Ratio

Index Number:

- b) If an investment cost is \$5,000 and the investment is expected to generate cash inflows as follows.

Year 1	\$ 1,500
Year 2	\$ 2,000
Year 3	\$ 2,000
Year 4	\$ 1,500

- i. What is the payback period for the investment?

[02 Marks]

- ii. If the above investment is allowed to depreciate on the straight line basis and the scrap value is \$1,000 at end of the 4th year, calculate the annual depreciation, average annual profit, and accounting rate of return assuming that there are no other expenses.

[04 Marks]

- Annual depreciation
- Average annual profit
- Accounting rate of return

- c) Amalka Ltd intends to purchase a suitable machine to increase the organization's production capacity. The following information was collected by the financial division of the Organization.

	Cash Inflows	
	Machine A	Machine B
Initial Investment	\$ (5,000)	\$ (5,000)
Year 1	\$ 1,000	\$ 1,500
Year 2	\$ 1,000	\$ 2,000
Year 3	\$ 2,500	\$ 2,500
Year 4	\$ 3,000	\$ 3,000

- i. If the company considers a 5% rate of return, calculate the Net Present Value for the two investments.

[06 Marks]

Index Number:

- ii. What is the most suitable machine based on Net Present Value (NPV)?
Justify your Answer.

[02 Marks]

- iii. Define "Time value of money".

[02 Marks]

Present Value Table

Periods (n)	Interest rates (r)									
	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467
9	0.914	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.350
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239
16	0.853	0.728	0.623	0.534	0.458	0.394	0.339	0.292	0.252	0.218
17	0.844	0.714	0.605	0.513	0.436	0.371	0.317	0.270	0.231	0.198
18	0.836	0.700	0.587	0.494	0.416	0.350	0.296	0.250	0.212	0.180
19	0.828	0.686	0.570	0.475	0.396	0.331	0.277	0.232	0.194	0.164
20	0.820	0.673	0.554	0.456	0.377	0.312	0.258	0.215	0.178	0.149
